

Relatório 2ª Aula Prática

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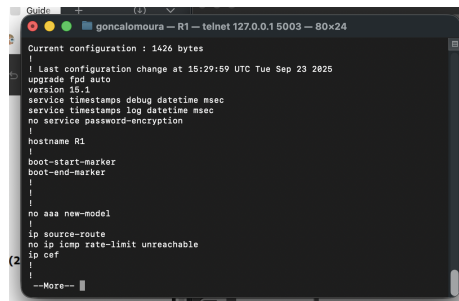
23 setembro 2025

1 Instalação

Foram seguidas as instruções de instalação do professor presentes no Discord referentes a alunos com computadores com sistema operativo macOS.

2 Configuração inicial

Foi utilizada a configuração de rede criada na aula passada. Acabámos por configurar o router e os dois respetivos computadores.



```
goncalomoura - R1 - telnet 127.0.0.1 5003 - 80x24
Current configuration : 1426 bytes
! Last configuration change at 15:29:59 UTC Tue Sep 23 2025
upgrade ftd auto
version 15.1
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
no aaa new-model
!
ip source-route
no ip icmp rate-limit unreachable
ip cef
(R1)
--More--
```

Figure 1: Configuração do Router

3 Periodicidade dos pacotes

Subtraindo o *request* ao *reply*, é possível obter o tempo que dura uma comunicação entre o router e o computador.

84 bytes from 192.168.1.1 icmp_seq=96 ttl=255 time=6.396 ms

No.	Time	Source	Destination	Protocol	Length	Info
36	25.733594	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
37	25.761572	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
38	25.845558	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
39	25.871100	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
40	26.773207	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
41	26.794962	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
42	27.844439	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
43	27.870822	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
44	27.780352	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
45	27.802275	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
46	28.874004	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
47	28.894890	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
48	29.846468	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
49	29.871843	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
50	29.871848	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
51	29.899919	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
52	29.875757	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request
53	29.892222	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) reply
54	30.891159	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request
55	30.912252	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) reply
56	30.830313	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request
57	31.111051	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) reply
58	31.115883	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request
59	31.870886	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) reply

Figure 2: Alguns pacotes capturados no Wireshark

4 Valores observados no Wireshark

- PC Ethernet/MAC address: 192.168.1.100 at 00:50:79:66:68:00
- Router Ethernet/MAC address: 192.168.1.1 at ca:01:74:d3:00:08
- IPv4 (0x0800)
- Protocol: ICMP(1) é 01, como pode ser visto na Figura 3.
- para o mesmo request o id do reply é o mesmo.

```

[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: ca:01:74:d3:00:08 (ca:01:74:d3:00:08), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: ca:01:74:d3:00:08 (ca:01:74:d3:00:08)
  Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.100
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 64
  Identification: 0xc68c (58828)
  > 0000 .... = Flags: 0x0
  ... 0 0000 0000 0000 = Fragment Offset: 0
  Time to live: 255
  Protocol: ICMP (1)
  Header Checksum: 0x7166 [validation disabled]
  (Header Checksum: 0x7166 [validation disabled])
  
```

Figure 3: Código hexadecimal ICMP observado no Wireshark

5 ARP

5.1 ARP Request

Sender MAC address: Private-66:68:00 (00:50:79:66:68:00)
 Sender IP address: 192.168.1.100
 Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 Target IP address: 192.168.1.1

5.2 ARP Response

Sender MAC address: ca:01:8e:a1:00:08 (ca:01:8e:a1:00:08)
 Sender IP address: 192.168.1.1
 Target MAC address: Private-66:68:00 (00:50:79:66:68:00)
 Target IP address: 192.168.1.100

6 Padding

Com o comando “ping 192.168.1.1 -l 8” observamos que apenas um packet teve padding.

[illegible]

Figure 4: Padding apenas num packet

Já com o comando `ping 192.168.1.200 size 36`, o padding ocorre em todos os Ethernet headers dos packets ICMP.

7 Fragmentation

- Quando um packet é maior do que 1500 bytes, este tem de ser fragmentado, de modo que cada fragmento tem no máximo 1500 bytes.
- Os fragmentos com o mesmo ID são agrupados, formando o packet completo.
- Por exemplo, um packet com tamanho 2028 vai ter um fragmento com 1500 bytes e outro com 528.

8 Routing

Um novo switch e VPC foi adicionado (VPC3).

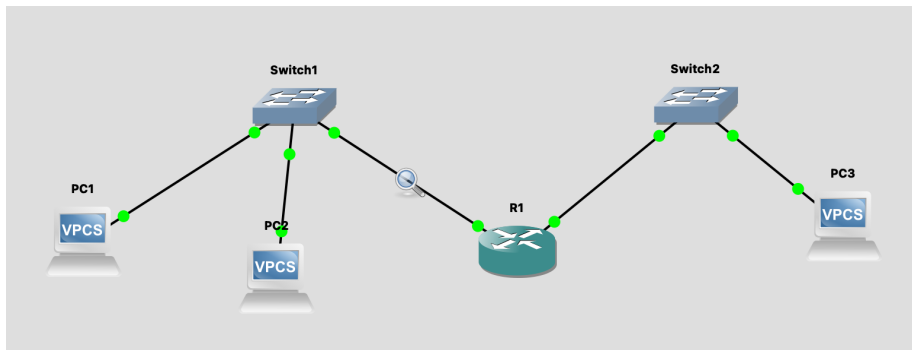


Figure 5: Network expandida

Nova tabela de Routing da Network:

```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, FastEthernet2/0
L    192.168.3.1/32 is directly connected, FastEthernet2/0

```

Figure 6: Tabela de Routing do Router

Antes da configuração dos gateways:

```

PC3> ping 192.168.1.100

No gateway found

```

Figure 7: ping falhado: No gateway found

Após configuração do gateway em ambos PC1 (gateway:192.168.1.1) e PC3(gateway 192.168.3.1):

```

PC1> ping 192.168.3.33

84 bytes from 192.168.3.33 icmp_seq=1 ttl=63 time=20.710 ms
84 bytes from 192.168.3.33 icmp_seq=2 ttl=63 time=23.627 ms
84 bytes from 192.168.3.33 icmp_seq=3 ttl=63 time=21.509 ms
84 bytes from 192.168.3.33 icmp_seq=4 ttl=63 time=23.882 ms
84 bytes from 192.168.3.33 icmp_seq=5 ttl=63 time=24.249 ms

```

Figure 8: Packets enviados do PC1 para o PC3

```

PC3> ping 192.168.1.100
84 bytes from 192.168.1.100 icmp_seq=1 ttl=63 time=178.370 ms
84 bytes from 192.168.1.100 icmp_seq=2 ttl=63 time=23.187 ms
84 bytes from 192.168.1.100 icmp_seq=3 ttl=63 time=17.249 ms
84 bytes from 192.168.1.100 icmp_seq=4 ttl=63 time=20.033 ms
84 bytes from 192.168.1.100 icmp_seq=5 ttl=63 time=24.345 ms

```

Figure 9: Packets enviados do PC3 para o PC1

icmp			
No.	Time	Source	Destination
7	29.629116	192.168.3.33	192.168.1.100
8	29.807188	192.168.1.100	192.168.3.33
10	30.810581	192.168.3.33	192.168.1.100
11	30.832505	192.168.1.100	192.168.3.33
12	31.833874	192.168.3.33	192.168.1.100
13	31.849849	192.168.1.100	192.168.3.33
14	32.851864	192.168.3.33	192.168.1.100

Figure 10: Packets enviados do PC3 para o PC1 observados no Wireshark

2.6

```
*echo Request pc1 -> r1
Source Mac Address: 00:50:79:66:68:00
destination Mac Address: ca:01:61:8d:00:08

source ip Address: 192.168.1.100
destination ip Address: 192.168.3.33

*reply r1 -> pc3
Source Mac Address: ca:01:61:8d:00:38
destination Mac Address: 00:50:79:66:68:02

source ip Address: 192.168.1.100
destination ip Address: 192.168.3.33

*request pc3 -> r1
Source Mac Address: 00:50:79:66:68:02
destination Mac Address: ca:01:61:8d:00:38

source ip Address: 192.168.1.33
destination ip Address: 192.168.3.100

*echo Reply r1 -> pc1
Source Mac Address: ca:01:61:8d:00:08
destination Mac Address: 00:50:79:66:68:00

source ip Address: 192.168.3.33
destination ip Address: 192.168.1.100
```

Figure 11: Comunicação entre PC1, Router e PC3

2.7

16	54.919206	192.168.3.33	192.168.1.10	ICMP	98 Echo (ping) request
17	56.920730	192.168.3.33	192.168.1.10	ICMP	98 Echo (ping) request
18	58.921854	192.168.3.33	192.168.1.10	ICMP	98 Echo (ping) request
19	60.922744	192.168.3.33	192.168.1.10	ICMP	98 Echo (ping) request
20	62.923867	192.168.3.33	192.168.1.10	ICMP	98 Echo (ping) request

Figure 12: Packets do ping de PC3 para 192.168.1.10

2.8

3	0.057851	192.168.3.33	194.100.1.1	ICMP	98 Echo (ping) request id=0x7234, seq=1/256, 1
4	0.067867	192.168.3.1	192.168.3.33	ICMP	70 Destination unreachable (Host unreachable)
5	1.070269	192.168.3.33	194.100.1.1	ICMP	98 Echo (ping) request id=0x7334, seq=2/512, 1
6	1.084982	192.168.3.1	192.168.3.33	ICMP	70 Destination unreachable (Host unreachable)
7	2.087504	192.168.3.33	194.100.1.1	ICMP	98 Echo (ping) request id=0x7434, seq=3/768, 1
8	2.101692	192.168.3.1	192.168.3.33	ICMP	70 Destination unreachable (Host unreachable)
9	3.103817	192.168.3.33	194.100.1.1	ICMP	98 Echo (ping) request id=0x7534, seq=4/1024, 1
10	3.116619	192.168.3.1	192.168.3.33	ICMP	70 Destination unreachable (Host unreachable)
11	4.118003	192.168.3.33	194.100.1.1	ICMP	98 Echo (ping) request id=0x7634, seq=5/1280, 1
12	4.125457	192.168.3.1	192.168.3.33	ICMP	70 Destination unreachable (Host unreachable)

Figure 13: Packets do ping de PC3 para 194.100.1.1

2.9

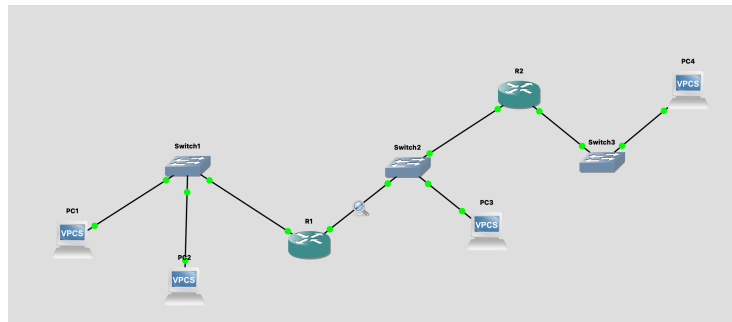


Figure 14: Network Atualizada

2.10

A nova routing table é igual à anterior.

	192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.1.0/24 is directly connected, GigabitEthernet0/0
L	192.168.1.1/32 is directly connected, GigabitEthernet0/0
	192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.3.0/24 is directly connected, FastEthernet2/0
L	192.168.3.1/32 is directly connected, FastEthernet2/0

Figure 15: Nova Routing Table de R1

2.11 Há apenas requests e não há respostas, visto que 192.168.3.150 não existe na network

8	52.584625	Private_66:68:00	Broadcast	ARP	64
9	52.686601	ca:01:8e:a1:00:08	Private_66:68:00	ARP	60
10	52.687929	192.168.1.100	192.168.3.150	ICMP	98
11	54.689783	192.168.1.100	192.168.3.150	ICMP	98
12	56.692019	192.168.1.100	192.168.3.150	ICMP	98
13	58.693777	192.168.1.100	192.168.3.150	ICMP	98
14	60.006425	ca:01:8e:a1:00:08	ca:01:8e:a1:00:08	LOOP	60
15	60.695655	192.168.1.100	192.168.3.150	ICMP	98

Figure 16: Packets do ping de PC1 para 192.168.3.150

2.12

6	18.446088	192.168.1.1	192.168.1.100	ICMP	70 Destination unreachable (Host unreachable)
7	19.449514	192.168.1.100	192.168.2.254	ICMP	98 Echo (ping) request id=0x5f39, seq=2/512, t
8	19.460198	192.168.1.1	192.168.1.100	ICMP	70 Destination unreachable (Host unreachable)
10	20.461486	192.168.1.100	192.168.2.254	ICMP	98 Echo (ping) request id=0x6039, seq=3/768, t
11	20.466654	192.168.1.1	192.168.1.100	ICMP	70 Destination unreachable (Host unreachable)
12	21.469631	192.168.1.100	192.168.2.254	ICMP	98 Echo (ping) request id=0x6139, seq=4/1024, i
13	21.483017	192.168.1.1	192.168.1.100	ICMP	70 Destination unreachable (Host unreachable)
14	22.488408	192.168.1.100	192.168.2.254	ICMP	98 Echo (ping) request id=0x6239, seq=5/1280, i
15	22.493576	192.168.1.1	192.168.1.100	ICMP	70 Destination unreachable (Host unreachable)

Figure 17: Packets do ping de PC1 para 192.168.2.254

2.13

```
Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 192.168.3.254
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, FastEthernet2/0
L    192.168.3.1/32 is directly connected, FastEthernet2/0
R1#
```

Figure 18: Routing Table R1

```
Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 192.168.3.1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0
L    192.168.2.254/32 is directly connected, GigabitEthernet0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, FastEthernet2/0
L    192.168.3.254/32 is directly connected, FastEthernet2/0
R2#
```

Figure 19: Routing Table R2

2.14

4	4.785688	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x0e3f, seq=1/256, ttl=1 (no response found!)
5	6.333853	192.168.1.108	192.168.2.22	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
6	6.333853	192.168.1.108	192.168.2.22	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
7	5.891186	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x0f3f, seq=2/512, ttl=1 (no response found!)
8	5.967445	192.168.1.1	192.168.1.108	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
9	6.965081	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x103f, seq=3/768, ttl=1 (no response found!)
10	6.973782	192.168.1.1	192.168.1.108	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
11	7.975142	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x113f, seq=4/1024, ttl=1 (no response found!)
12	7.989758	192.168.1.1	192.168.1.108	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
13	8.991323	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x123f, seq=5/1280, ttl=1 (no response found!)
14	8.996944	192.168.1.1	192.168.1.108	ICMP	70 Time-to-live exceeded (Time to live exceeded in transit)
16	13.457443	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x173f, seq=1/256, ttl=2 (no response found!)
17	15.458752	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x103f, seq=2/512, ttl=2 (no response found!)
18	17.460193	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x1b3f, seq=3/768, ttl=2 (no response found!)
19	19.461218	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x1d3f, seq=4/1024, ttl=2 (no response found!)
21	21.463488	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x1f3f, seq=5/1280, ttl=2 (no response found!)
22	25.541552	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x233f, seq=1/256, ttl=3 (no response found!)
23	27.543778	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x253f, seq=2/512, ttl=3 (no response found!)
24	29.545569	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x273f, seq=3/768, ttl=3 (no response found!)
26	31.546816	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x293f, seq=4/1024, ttl=3 (no response found!)
27	33.547997	192.168.1.108	192.168.2.22	ICMP	98 Echo (ping) request id=0x2b3f, seq=5/1280, ttl=3 (no response found!)

Figure 20: Packets de Pings com TTL diferentes.